CSE370 Lab Assignment - 02

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Section: 04

MySQL commands:

- mysql -u root -p
- “”
- show databases;
- create database **CSE370_lab_assignment_2**;
- use **CSE370_lab_assignment_2**;

Creating table:

- create table employees(

  employee_id char(10),

  first_name varchar(20),

  last_name varchar(20),

  email varchar(60),

  phone_number char(14),

  hire_date date,

```
job_id char(10),

salary int,

commission_pct decimal(5,3),

manager_id char(10),

department_id char(10)

);
```

```
MariaDB [(none)]> use cse370_lab_assignment_2;
Database changed
MariaDB [cse370_lab_assignment_2]> create table employees(
    -> employee_id char(10),
    -> first_name varchar(20),
    -> last_name varchar(20),
    -> email varchar(60),
    -> phone_number char(14),
    -> hire_date date,
    -> job_id char(10),
    -> salary int,
    -> commission_pct decimal(5,3),
    -> manager_id char(10),
    -> department_id char(10)
    -> );
Query OK, 0 rows affected (0.393 sec)

MariaDB [cse370_lab_assignment_2]> describe employees;
+----------------+--------------+------+-----+---------+-------+
| Field          | Type         | Null | Key | Default | Extra |
+----------------+--------------+------+-----+---------+-------+
| employee_id    | char(10)     | YES  |     | NULL    |       |
| first_name     | varchar(20)  | YES  |     | NULL    |       |
| last_name      | varchar(20)  | YES  |     | NULL    |       |
| email          | varchar(60)  | YES  |     | NULL    |       |
| phone_number   | char(14)     | YES  |     | NULL    |       |
| hire_date      | date         | YES  |     | NULL    |       |
| job_id         | char(10)     | YES  |     | NULL    |       |
| salary         | int(11)      | YES  |     | NULL    |       |
| commission_pct | decimal(5,3) | YES  |     | NULL    |       |
| manager_id     | char(10)     | YES  |     | NULL    |       |
| department_id  | char(10)     | YES  |     | NULL    |       |
+----------------+--------------+------+-----+---------+-------+
11 rows in set (0.472 sec)
```

Inserting into Table:

- insert into Employees values

  ('EMP001','James','Gosling','james@gmail.com','+19123446789
  87','2021-2-28','JOB001',29000,37.600,'MNG001','DPT001'),

  ('EMP002','Guido','Rossum','guido@outlook.com','+1988843155
  021','2021-4-02','JOB002',29000,43.150,'MNG001','DPT002'),

  ('EMP003','Brendan','Eich','eich@hotmail.com','+19200554866
  20','2021-6-02','JOB005',48000,31.760,'MNG003','DPT005'),

  ('EMP004','Mangal','Pandey','pandey@gmail.com','+1915514863
  255','2021-5-01','JOB003',55000,55.600,'MNG002','DPT003'),

  ('EMP005','Ryan','Dahl','ryan@msn.com','+1946521032598','20
  21-9-05','JOB004',25000,25.00,'MNG002','DPT004'),

  ('EMP006','Rasmus','Lerdorf','lerdorf@gmail.com','+19557403
  94251','2023-3-01','JOB010',28000,71.050,'MNG003','DPT005')
  ,

  ('EMP007','Satoshi','Nakamoto','nakamoto@yahoo.com','+19440
  31854001','2023-9-05','JOB010',70000,20.000,'MNG004','DPT00
  7'),

  ('EMP008','Simon','Willison','simon@yahoo.com','+1953184544
  992','2023-9-05','JOB009',70000,15.860,'MNG004','DPT007'),

  ('EMP009','Taylor','Otwell','otwell@gmail.com','+1914532001
  895','2023-6-02','JOB008',32000,28.310,'MNG003','DPT006'),

  ('EMP010','Evan','You','you@outlook.com','+1914148445660','
  2021-7-04','JOB007',51000,23.260,'MNG003','DPT003'),

  ('EMP011','Adrian','Holovaty','adrian@gmail.com','+19551244
  62514','2023-9-05','JOB005',34000,33.220,'MNG004','DPT007')
  ,

```
('EMP012','Rod','Johnson','rod@outlook.com','+1925874136901
','2023-5-03','JOB011',58000,45.670,'MNG004','DPT007');
```

```
MariaDB [cse370_lab_assignment_2]> insert into Employees values
    -> ('EMP001','James','Gosling','james@gmail.com','+1912344678987','2021-2-28','JOB001',29000,37.600,'MNG001','DPT001'),
    -> ('EMP002','Guido','Rossum','guido@outlook.com','+1988843155021','2021-4-02','JOB002',29000,43.150,'MNG001','DPT002'),
    -> ('EMP003','Brendan','Eich','eich@hotmail.com','+1920055486620','2021-6-02','JOB005',48000,31.760,'MNG003','DPT005'),
    -> ('EMP004','Mangal','Pandey','pandey@gmail.com','+1915514863255','2021-5-01','JOB003',55000,55.600,'MNG002','DPT003'),
    -> ('EMP005','Ryan','Dahl','ryan@msn.com','+1946521032598','2021-9-05','JOB004',25000,25.00,'MNG002','DPT004'),
    -> ('EMP006','Rasmus','Lerdorf','lerdorf@gmail.com','+1955740394251','2023-3-01','JOB010',28000,71.050,'MNG003','DPT005'),
    -> ('EMP007','Satoshi','Nakamoto','nakamoto@yahoo.com','+1944031854001','2023-9-05','JOB010',70000,20.000,'MNG004','DPT007'),
    -> ('EMP008','Simon','Willison','simon@yahoo.com','+1953184544992','2023-9-05','JOB009',70000,15.860,'MNG004','DPT007'),
    -> ('EMP009','Taylor','Otwell','otwell@gmail.com','+1914532001895','2023-6-02','JOB008',32000,28.310,'MNG003','DPT006'),
    -> ('EMP010','Evan','You','you@outlook.com','+1914148445660','2021-7-04','JOB007',51000,23.260,'MNG003','DPT003'),
    -> ('EMP011','Adrian','Holovaty','adrian@gmail.com','+1955124462514','2023-9-05','JOB005',34000,33.220,'MNG004','DPT007'),
    -> ('EMP012','Rod','Johnson','rod@outlook.com','+1925874136901','2023-5-03','JOB011',58000,45.670,'MNG004','DPT007');
Query OK, 12 rows affected (0.136 sec)
Records: 12  Duplicates: 0  Warnings: 0

MariaDB [cse370_lab_assignment_2]> select * from employees;
+-------------+------------+-----------+--------------------+----------------+------------+--------+--------+----------------+------------+---------------+
| employee_id | first_name | last_name | email              | phone_number   | hire_date  | job_id | salary | commission_pct | manager_id | department_id |
+-------------+------------+-----------+--------------------+----------------+------------+--------+--------+----------------+------------+---------------+
| EMP001      | James      | Gosling   | james@gmail.com    | +1912344678987 | 2021-02-28 | JOB001 |  29000 |         37.600 | MNG001     | DPT001        |
| EMP002      | Guido      | Rossum    | guido@outlook.com  | +1988843155021 | 2021-04-02 | JOB002 |  29000 |         43.150 | MNG001     | DPT002        |
| EMP003      | Brendan    | Eich      | eich@hotmail.com   | +1920055486620 | 2021-06-02 | JOB005 |  48000 |         31.760 | MNG003     | DPT005        |
| EMP004      | Mangal     | Pandey    | pandey@gmail.com   | +1915514863255 | 2021-05-01 | JOB003 |  55000 |         55.600 | MNG002     | DPT003        |
| EMP005      | Ryan       | Dahl      | ryan@msn.com       | +1946521032598 | 2021-09-05 | JOB004 |  25000 |         25.000 | MNG002     | DPT004        |
| EMP006      | Rasmus     | Lerdorf   | lerdorf@gmail.com  | +1955740394251 | 2023-03-01 | JOB010 |  28000 |         71.050 | MNG003     | DPT005        |
| EMP007      | Satoshi    | Nakamoto  | nakamoto@yahoo.com | +1944031854001 | 2023-09-05 | JOB010 |  70000 |         20.000 | MNG004     | DPT007        |
| EMP008      | Simon      | Willison  | simon@yahoo.com    | +1953184544992 | 2023-09-05 | JOB009 |  70000 |         15.860 | MNG004     | DPT007        |
| EMP009      | Taylor     | Otwell    | otwell@gmail.com   | +1914532001895 | 2023-06-02 | JOB008 |  32000 |         28.310 | MNG003     | DPT006        |
| EMP010      | Evan       | You       | you@outlook.com    | +1914148445660 | 2021-07-04 | JOB007 |  51000 |         23.260 | MNG003     | DPT003        |
| EMP011      | Adrian     | Holovaty  | adrian@gmail.com   | +1955124462514 | 2023-09-05 | JOB005 |  34000 |         33.220 | MNG004     | DPT007        |
| EMP012      | Rod        | Johnson   | rod@outlook.com    | +1925874136901 | 2023-05-03 | JOB011 |  58000 |         45.670 | MNG004     | DPT007        |
+-------------+------------+-----------+--------------------+----------------+------------+--------+--------+----------------+------------+---------------+
12 rows in set (0.000 sec)
```

Task-1:

Find the **first_name, last_name, email, phone_number, hire_date** and **department_id** of all the employees with the latest **hire_date**.

- select **first_name, last_name, email, phone_number, hire_date, department_id** from employees

  where **hire_date** = (select max(**hire_date**) from employees);

```
MariaDB [cse370_lab_assignment_2]> select first_name, last_name, email, phone_number, hire_date, department_id from employees
    -> where hire_date = (select max(hire_date) from employees);
+------------+-----------+--------------------+----------------+------------+---------------+
| first_name | last_name | email              | phone_number   | hire_date  | department_id |
+------------+-----------+--------------------+----------------+------------+---------------+
| Satoshi    | Nakamoto  | nakamoto@yahoo.com | +1944031854001 | 2023-09-05 | DPT007        |
| Simon      | Willison  | simon@yahoo.com    | +1953184544992 | 2023-09-05 | DPT007        |
| Adrian     | Holovaty  | adrian@gmail.com   | +1955124462514 | 2023-09-05 | DPT007        |
+------------+-----------+--------------------+----------------+------------+---------------+
3 rows in set (0.042 sec)
```

Task-2:

Find the ***first_name***, ***last_name***, ***employee_id***, ***phone_number, salary*** and ***department_id*** of all the employees with the lowest **salary** in each department.

- select **first_name, last_name, employee_id, phone_number, salary, department_id** from employees g1

  where g1.**salary** = (select min(**salary**) from employees g2

  where g1.**department_id** = g2.**department_id**);

```
MariaDB [cse370_lab_assignment_2]> select first_name, last_name, employee_id, phone_number, salary, department_id from employees g1
    -> where g1.salary = (select min(salary) from employees g2
    -> where g1.department_id = g2.department_id);
+------------+-----------+-------------+----------------+--------+---------------+
| first_name | last_name | employee_id | phone_number   | salary | department_id |
+------------+-----------+-------------+----------------+--------+---------------+
| James      | Gosling   | EMP001      | +1912344678987 |  29000 | DPT001        |
| Guido      | Rossum    | EMP002      | +1988843155021 |  29000 | DPT002        |
| Ryan       | Dahl      | EMP005      | +1946521032598 |  25000 | DPT004        |
| Rasmus     | Lerdorf   | EMP006      | +1955740394251 |  28000 | DPT005        |
| Taylor     | Otwell    | EMP009      | +1914532001895 |  32000 | DPT006        |
| Evan       | You       | EMP010      | +1914148445660 |  51000 | DPT003        |
| Adrian     | Holovaty  | EMP011      | +1955124462514 |  34000 | DPT007        |
+------------+-----------+-------------+----------------+--------+---------------+
7 rows in set (0.781 sec)
```

Task-3:

Find the ***first_name***, ***last_name***, ***employee_id***, ***commission_pct*** *and* ***department_id*** of all the employees in the department 'DPT007' who have a lower commission_pct than all of the employees of the department 'DPT005'.

- select **first_name, last_name, employee_id, commission_pct, department_id** from employees

  where **department_id** = 'DPT007' and **commission_pct** < all(select **commission_pct** from employees

```
where department_id = 'DPT005');
```

```
MariaDB [cse370_lab_assignment_2]> select first_name, last_name, employee_id, commission_pct, department_id from employees
    -> where department_id = 'DPT007' and commission_pct < all(select commission_pct from employees
    -> where department_id = 'DPT005');
+------------+-----------+-------------+----------------+---------------+
| first_name | last_name | employee_id | commission_pct | department_id |
+------------+-----------+-------------+----------------+---------------+
| Satoshi    | Nakamoto  | EMP007      |         20.000 | DPT007        |
| Simon      | Willison  | EMP008      |         15.860 | DPT007        |
+------------+-----------+-------------+----------------+---------------+
2 rows in set (0.038 sec)
```

Task-4:

Find the **department_id** and total number of employees of each department which does not have a single employee under it with a **salary** more than 30,000.

- `select` **`department_id`**`, count(`**`employee_id`**`) as` **`total_employees`** `from employees`

  `group BY` **`department_id`** `having max(`**`salary`**`) <= 30000;`

```
MariaDB [cse370_lab_assignment_2]> select department_id, count(employee_id) as total_employees from employees
    -> group BY department_id having max(salary) <= 30000;
+---------------+-----------------+
| department_id | total_employees |
+---------------+-----------------+
| DPT001        |               1 |
| DPT002        |               1 |
| DPT004        |               1 |
+---------------+-----------------+
3 rows in set (0.001 sec)
```

Task-5:

For each of the departments, find the ***department_id***, ***job_id*** and ***commission_pct*** with ***commission_pct*** less than at least one other ***job_id*** in that department.

- select **department_id, job_id, commission_pct** from employees g1

  where g1.**commission_pct** < any (select g2.**commission_pct** from employees g2

  where g2.**department_id** = g1.**department_id** and g2.**job_id** != g1.**job_id**);

```
MariaDB [cse370_lab_assignment_2]> select department_id, job_id, commission_pct from employees g1
    -> where g1.commission_pct < any (select g2.commission_pct from employees g2
    -> where g2.department_id = g1.department_id and g2.job_id != g1.job_id);
+---------------+--------+----------------+
| department_id | job_id | commission_pct |
+---------------+--------+----------------+
| DPT005        | JOB005 |         31.760 |
| DPT007        | JOB010 |         20.000 |
| DPT007        | JOB009 |         15.860 |
| DPT003        | JOB007 |         23.260 |
| DPT007        | JOB005 |         33.220 |
+---------------+--------+----------------+
5 rows in set (0.213 sec)
```

Task-6:

1. Find the ***manager_id*** who does not have any employee under them with a ***salary*** less than 3500.

   - select distinct **manager_id** from employees where **salary** > 3500;

```
MariaDB [cse370_lab_assignment_2]> select distinct manager_id from employees where salary > 3500;
+------------+
| manager_id |
+------------+
| MNG001     |
| MNG003     |
| MNG002     |
| MNG004     |
+------------+
4 rows in set (0.773 sec)
```

Task-7:

Find the ***first_name***, ***last_name***, ***employee_id***, ***email, salary, department_id*** *and* ***commission_pct*** of the employee who has the lowest ***commission_pct*** under each manager.

- select **first_name, last_name, employee_id, email, salary, department_id, commission_pct** from employees where (**manager_id, commission_pct**) in (select **manager_id,** min(**commission_pct**) from employees group by **manager_id**);

```
MariaDB [cse370_lab_assignment_2]> select first_name, last_name, employee_id, email, salary, department_id, commission_pct from employees
    -> where (manager_id, commission_pct) in (select manager_id, min(commission_pct) from employees
    -> group by manager_id);
+------------+-----------+-------------+-----------------+--------+---------------+----------------+
| first_name | last_name | employee_id | email           | salary | department_id | commission_pct |
+------------+-----------+-------------+-----------------+--------+---------------+----------------+
| James      | Gosling   | EMP001      | james@gmail.com |  29000 | DPT001        |         37.600 |
| Ryan       | Dahl      | EMP005      | ryan@msn.com    |  25000 | DPT004        |         25.000 |
| Simon      | Willison  | EMP008      | simon@yahoo.com |  70000 | DPT007        |         15.860 |
| Evan       | You       | EMP010      | you@outlook.com |  51000 | DPT003        |         23.260 |
+------------+-----------+-------------+-----------------+--------+---------------+----------------+
4 rows in set (0.001 sec)
```